DATA ANALYTICS WITH POWER BI

PROJECT REPORT

Interactive Power BI Cyber Attack Dashboard of World-wide

Submitted by:

Arpit Kumar

Registration No:

12312286

Programme and Section:

BTECH CSE K23CD

Course Code: INT374

Under the Guidance of

Dr. Madhu Bala

Discipline of CSE/IT

LOVELY PROFESSIONAL UNIVERSITY

Transforming Education Transforming India

Declaration

I, Arpit, student of BTECH in Computer Science and Engineering under CSE/IT Discipline at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20-12-2025

Registration No: 12312286 Arpit

CERTIFICATE

This is to certify that Arpit, bearing Registration no. 12303458 has completed INT374, project titled, **"Cyber Attack Dashboard"** under my guidance and supervision. To the best of my knowledge, the present work is the result of his/her original development, effort and study.

Signature and Name of the Supervisor

Designation of the Supervisor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 20-12-2025

Acknowledgement

I would like to express my heartfelt gratitude to everyone who supported and guided me throughout the course of this project titled “Interactive Cyber Attack Dashboard Dashboard.”

I am sincerely thankful to my faculty Dr. Madhu Bala for their continuous encouragement, valuable feedback, and constructive suggestions. Their insightful guidance helped me stay focused and improve the quality of my work at every stage.

This project has not only enhanced my technical proficiency in Power BI and data visualization but also strengthened my ability to interpret and communicate data-driven insights effectively.

Arpit

12312286

Table of Contents

1.	Introduction	6
2.	Source of Dataset	7
3.	Dataset Preprocessing	8
4.	Analysis on Dataset (for each objective)	9
5.	Visualization of Charts and Graphs	10-17
6.	Dashboard	18
7.	LinkedIn	19-20
8.	Conclusion	21
9.	Future Scope	22
10.	References	23

Introduction

With the rapid expansion of digital technologies, organizations across the globe are increasingly dependent on information systems, cloud services, and online platforms. While this digital transformation has improved efficiency and connectivity, it has also led to a significant rise in cyber threats. Cyberattacks such as malware infections, phishing campaigns, ransomware, SQL injection, and zero-day vulnerabilities pose serious risks to data confidentiality, system availability, and financial stability. As cybercriminal techniques continue to evolve, there is a growing need for data-driven approaches to understand attack patterns and mitigate their impact effectively.

This project presents a **Cyber Attack Dashboard** developed using **Power BI**, aimed at providing a comprehensive and interactive analysis of cyber threats across multiple dimensions. The dashboard visualizes key metrics such as total financial losses, number of affected users, attack frequency, vulnerability types, defense mechanisms, target industries, and geographical distribution. By integrating multiple filters and slicers, users can dynamically explore the data by country, year, attack type, target industry, and attack source, enabling deeper insight and root cause analysis.

Advanced **DAX calculations** have been implemented to derive meaningful KPIs, including average financial loss, peak attack hours, and attack severity percentages. Various visual elements such as cards, bar charts, donut charts, line graphs, treemaps, and geo-maps are used to present complex cybersecurity data in an intuitive and visually engaging manner. The second page of the dashboard focuses on detailed cyber threat analysis, highlighting industry-wise attack distribution, vulnerability exposure, effectiveness of defense mechanisms, and global attack trends.

Overall, this dashboard serves as a powerful analytical tool for cybersecurity professionals, researchers, and decision-makers. It supports informed decision-making by identifying high-risk industries, common attack vectors, and critical security gaps, ultimately contributing to improved cyber resilience and strategic security planning.

Source of the Dataset

The dataset used in this project represents **global cybersecurity threat data spanning from 2015 to 2024**. It is a structured and analytical dataset designed to capture key aspects of cyberattacks across different regions, industries, and time periods. The data includes information related to **attack types**, **target industries**, **security vulnerabilities**, **defense mechanisms**, **attack sources**, **financial losses**, and **affected users**, enabling a comprehensive analysis of cyber threat patterns.

This dataset is taken from opendata.

Here are the **column names highlighted clearly in a list**, exactly as present in your dataset:

- **Country**
- **Year**
- **Attack_ID**
- **Target Industry**
- **Financial Loss in Million Doller**
- **Number of Affected Users**
- **Attack Source**
- **Security Vulnerability Type**
- **Defense Mechanism Used**
- **Incident Resolution Time (in Hours)**

Each record in the dataset provides details such as the **year of attack**, **country or region affected**, **type of cyberattack** (e.g., malware, phishing, ransomware, SQL injection, DDoS, man-in-the-middle), and the **target industry** including IT, banking, healthcare, retail, government, and telecommunications. The dataset also captures **security vulnerability types** like weak passwords, unpatched software, social engineering, and zero-day vulnerabilities, along with the **defense mechanisms** used, such as antivirus, firewalls, VPNs, encryption, and Additionally,thedatasetquantifiesthe**impactofcyberattacks**byrecording**total losses**, **average loss values**, **number of affected users**, and **peak attack hours**. This enables trend analysis, risk assessment, and comparison across years and industries. Overall, the dataset is well-suited for visualization and analytical purposes, making it ideal for building interactive dashboards, performing root cause analysis, and supporting data-driven cybersecurity decision-making

Dataset Preprocessing

1- **Data Cleaning**

- Removed blank and null values from important columns such as *Country*, *Attack Type*, *Target Industry*, and *Year* to avoid incorrect visualizations.
- Corrected inconsistent spellings and naming formats (for example, country names and attack types) to maintain data uniformity.

2- **Data Type Conversion**

- Converted columns like *Year* to **Whole Number**, *Financial Loss* to **Decimal Number**, and *Incident Resolution Time* to **Whole Number (hours)** for accurate calculations and aggregations.
- Ensured text fields such as *Attack Source* and *Defense Mechanism Used* were set to **Text** data type.

3- **Handling Missing Values**

- Replaced missing numerical values (such as financial loss or affected users) with appropriate default values or removed incomplete records where necessary.
- Verified that missing values did not impact KPI cards and trend charts.

4- **Column Renaming & Formatting**

- Renamed column headers to more readable and professional names (e.g., "Financial Loss in Million Dollar") for better understanding in reports.
- Applied proper formatting to numerical columns, including currency symbols and thousand separators.

5- **Removing Duplicates**

- Checked for and removed duplicate *Attack_ID* entries to ensure each cyber incident was counted only once.

6- **Creating Calculated Columns & Measures**

- Created calculated columns and **DAX measures** such as *Total Financial Loss*, *Average Loss per Attack*, and *Total Affected Users* for analysis.

- Used aggregation functions to support KPIs and comparative visuals.

Analysis of Dataset

1. **To analyze the overall impact of cyberattacks globally** by visualizing total financial losses and the number of affected users across different years and regions.
2. **To identify the most common types of cyberattacks** such as malware, phishing, ransomware, and DDoS attacks using comparative bar and donut charts.
3. **To examine industry-wise vulnerability** by analyzing which target industries (IT, banking, healthcare, retail, government, etc.) are most frequently affected by cyber incidents.
4. **To evaluate security vulnerability patterns** by identifying common weaknesses such as weak passwords, unpatched software, and social engineering techniques.
5. **To assess the effectiveness of different defense mechanisms** including firewalls, antivirus systems, encryption, VPNs, and AI-based security solutions.
6. **To analyze geographical distribution of cyberattacks** using map visuals to identify high-risk countries and regions.
7. **To study yearly trends in cyberattacks** and understand how the frequency and financial impact of attacks have evolved over time.
8. **To measure incident resolution efficiency** by analyzing incident resolution time and identifying delays in response.
9. **To compare attack sources** (internal vs external threats) to understand the origin of cyber incidents.
10. **To enable interactive and dynamic analysis** through the use of slicers, filters, and drill-down features for deeper exploration.
11. **To support data-driven decision-making** by presenting insights in an intuitive and visually structured dashboard format.
12. **To demonstrate the use of Power BI analytical features** such as DAX measures, KPIs, and interactive visuals for cybersecurity analytics.

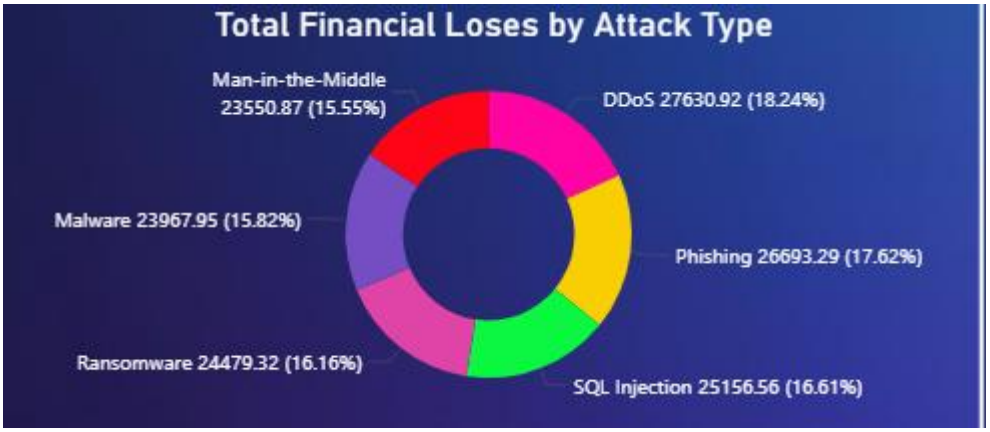
Visualization of Charts and Graphs

1. Total Financial Losses by Country (Top-Left Bar Chart)

This chart compares total financial losses caused by cyber, attacks across different countries. It helps identify regions most affected economically by cyber threats, enabling analysts to prioritize security strategies, allocate resources effectively, and understand geographical risk exposure based on attack impact.

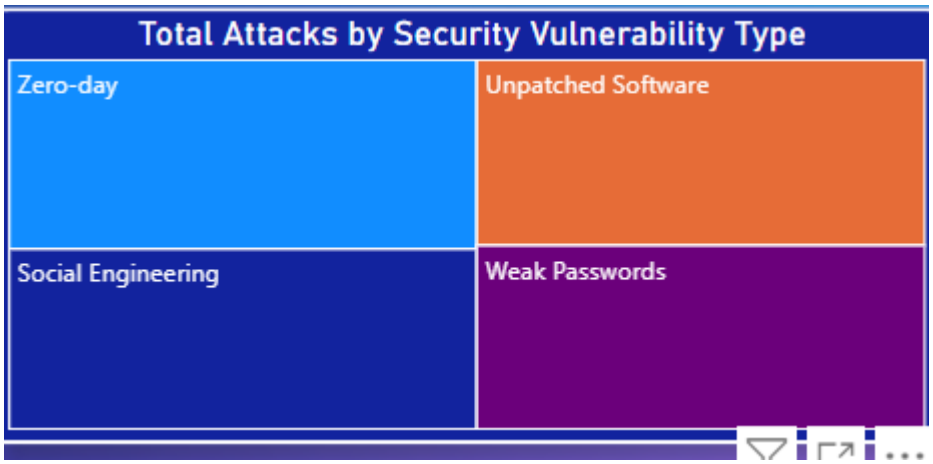
2. Total Financial Losses by Attack Type (Top-Centre Donut Chart)

This chart shows how total financial losses are distributed among different cyber, attack types such as phishing, malware, ransomware, DDoS, and SQL injection. It highlights which attack methods are most costly, helping organizations focus prevention efforts on the most financially damaging threats.



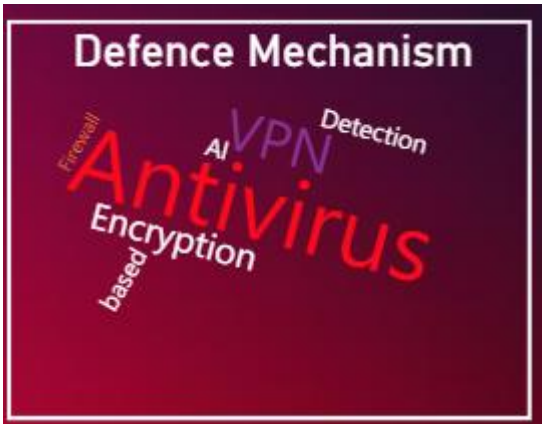
3. Total Attacks by Security Vulnerability Type (Top-Right Tree Map)

This visualization represents the number of attacks exploiting various security vulnerabilities like zero-day flaws, weak passwords, unpatched software, and social engineering. It helps identify the most commonly exploited weaknesses so organizations can strengthen controls and reduce exposure to frequent attack vectors.



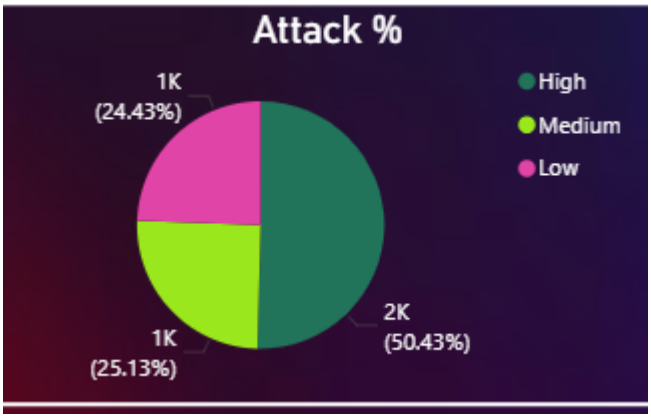
4. Defence Mechanism (Middle-Left Word Cloud)

This chart displays commonly used defense mechanisms such as antivirus, firewall, VPN, encryption, and intrusion detection. Larger words indicate more frequently used controls, providing insight into current security practices and showing which protective measures organizations rely on most to mitigate cyber threats.



5. Attack % by Severity Level (Middle Pie Chart)

This pie chart illustrates the proportion of cyber, attacks classified as high, medium, or low severity. It helps assess the overall threat intensity, showing how many attacks pose serious risks versus minor incidents, which supports better incident response planning and risk prioritization.



6. Industry and Year Analysis (Middle Table)

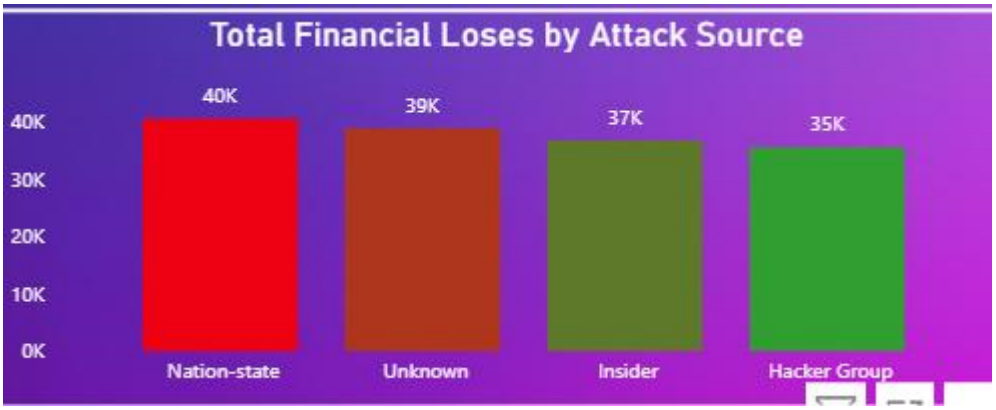
This table compares financial losses across industries over multiple years. It helps track trends, identify industries most targeted by cyber,attacks, and understand how financial impact evolves over time, enabling sector-specific risk assessment and informed cybersecurity investment decisions.

Industry and Year Analysis

Target Industry	2015	2016	2017	2018	2019	2020	2021
Telecommunications	1.85K	2.12K	1.63K	1.49K	2.14K	1.97K	2.73K
Retail	2.56K	1.73K	2.28K	2.10K	1.23K	2.14K	2.13K
IT	3.00K	2.07K	2.55K	1.94K	2.09K	2.74K	2.44K
Healthcare	1.64K	2.23K	2.70K	2.24K	1.72K	2.05K	2.11K
Government	1.78K	1.98K	2.75K	1.85K	2.19K	1.88K	1.97K
Total	**14.51K**	**13.95K**	**16.26K**	**14.72K**	**13.13K**	**15.77K**	**15.87K**

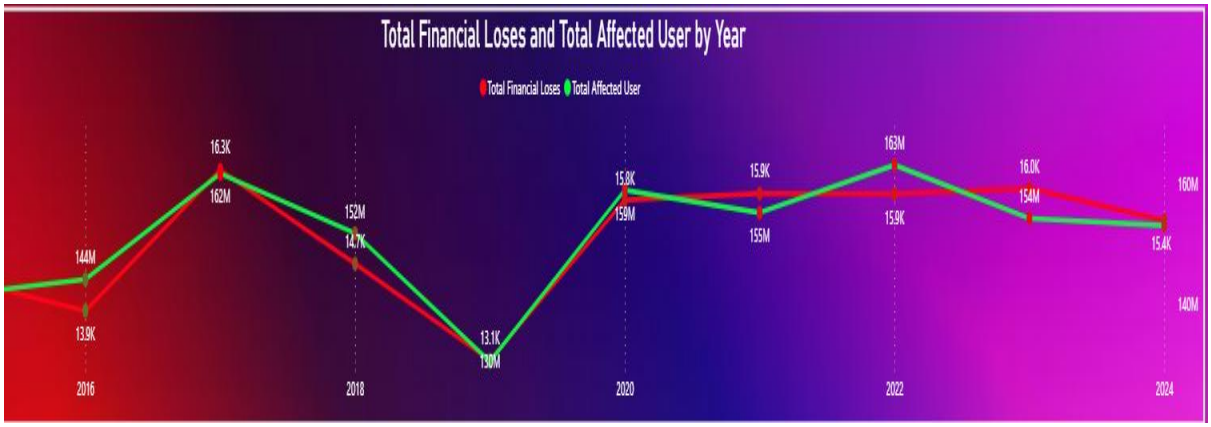
7. Total Financial Losses by Attack Source (Middle-Right Bar Chart)

This chart analyses financial losses based on attack sources such as nation-states, hacker groups, insiders, and unknown actors. It helps identify which threat actors cause the greatest damage and supports strategic planning for defending against specific attacker profiles.



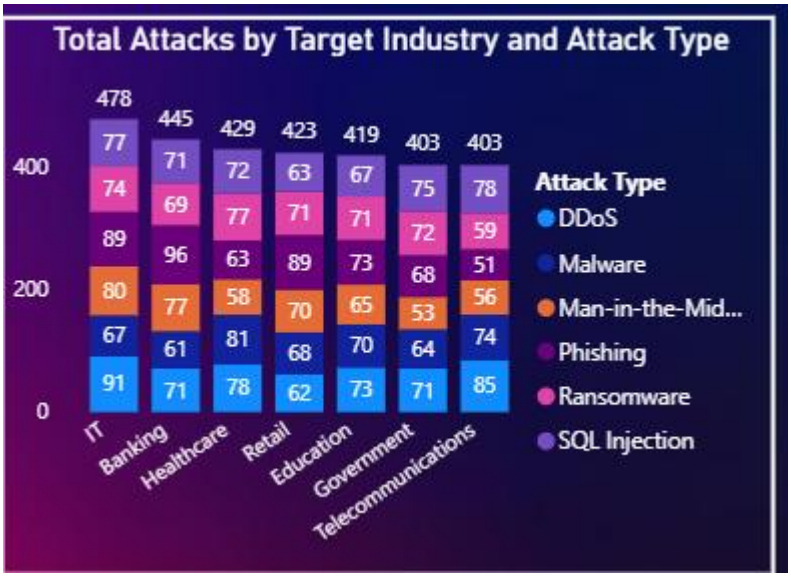
8. Total Financial Losses and Total Affected Users by Year (Bottom Line Chart)

This line chart tracks yearly trends in financial losses and affected users. It shows how cyber attack impact changes over time, helping stakeholders evaluate whether cybersecurity measures are improving, worsening, or remaining consistent across different years.



9.Total Attacks by Target Industry and Attack Type (Top-Left Stacked Bar Chart)

This chart compares the number of cyber, attacks across different target industries such as banking, healthcare, IT, and telecommunications, segmented by attack types. It helps identify which industries are most targeted and which attack techniques are most commonly used against each sector.



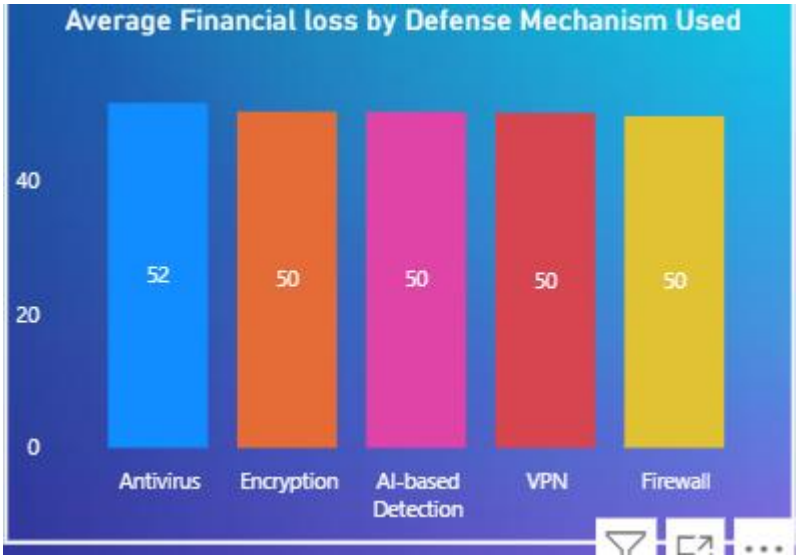
10.**World Cyber Attack Analysis (Top-Centre Map)**

This map visualizes global cyber, attacks geographically, showing attack distribution across countries. It helps identify global hotspots, understand regional threat concentration, and

analyze how cyber risks vary worldwide, supporting location-based security planning and international threat awareness.

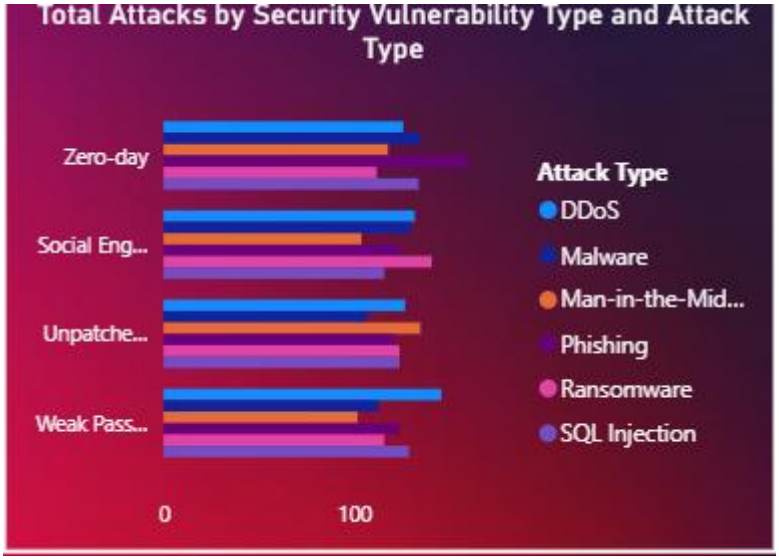
11. Average Financial Loss by Defense Mechanism Used (Top-Right Bar Chart)

This chart shows the average financial loss associated with different defense mechanisms like antivirus, encryption, VPNs, firewalls, and AI-based detection. It helps evaluate the effectiveness of security controls by comparing losses even after defences are implemented.



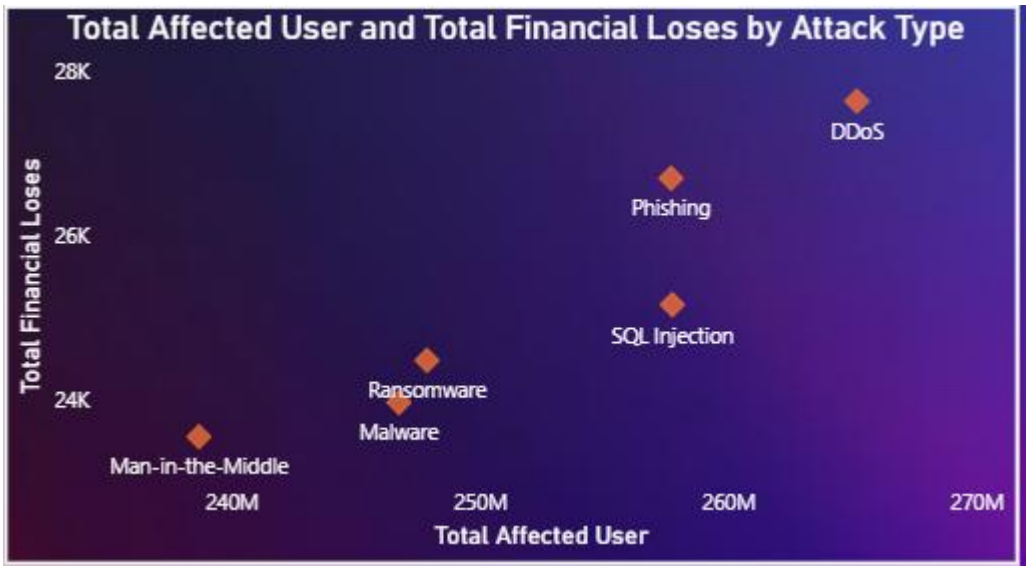
12. Total Attacks by Security Vulnerability Type and Attack Type (Middle-Left Bar Chart)

This visualization compares attack counts across vulnerabilities such as zero-day, social engineering, unpatched software, and weak passwords, further broken down by attack type. It highlights which vulnerabilities are most exploited and which attack methods dominate each weakness.



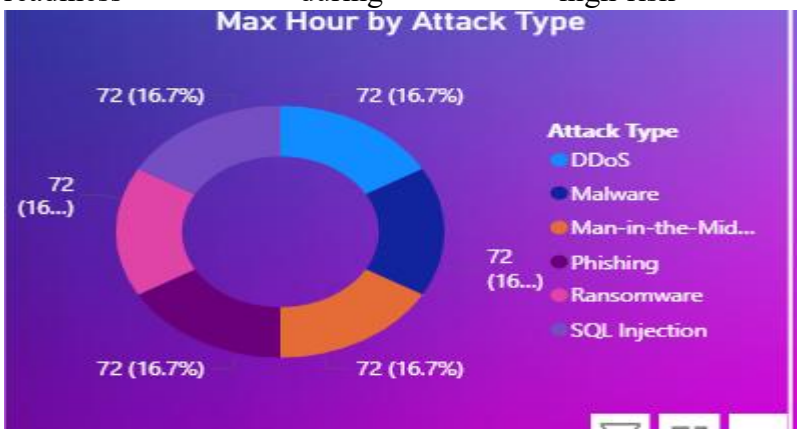
13. Total Affected Users and Total Financial Losses by Attack Type (Middle-Center Scatter Plot)

This chart plots attack types based on affected users versus financial losses. It helps identify attacks that cause widespread impact versus high financial damage, supporting prioritization of threats that are either costly, widespread, or both.



14. Max Hour by Attack Type (Middle-Right Donut Chart)

This donut chart shows peak attack hours for each attack type. It helps identify when attacks most frequently occur, enabling organizations to strengthen monitoring, staffing, and response readiness during high-risk time windows.

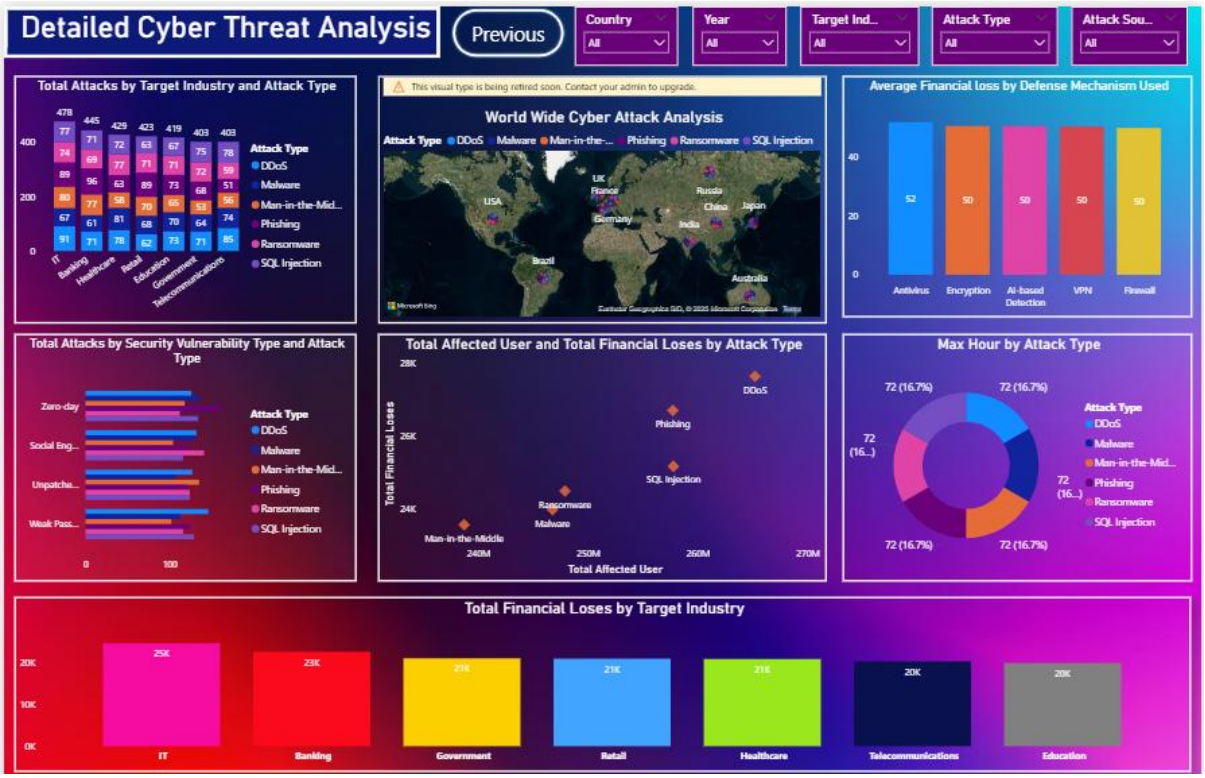


15. Total Financial Losses by Target Industry (Bottom Bar Chart)

This chart compares total financial losses across industries such as IT, banking, healthcare, retail, and education. It highlights sectors with the highest economic impact from cyber, attacks, helping stakeholders focus cybersecurity investments on the most financially vulnerable industries.

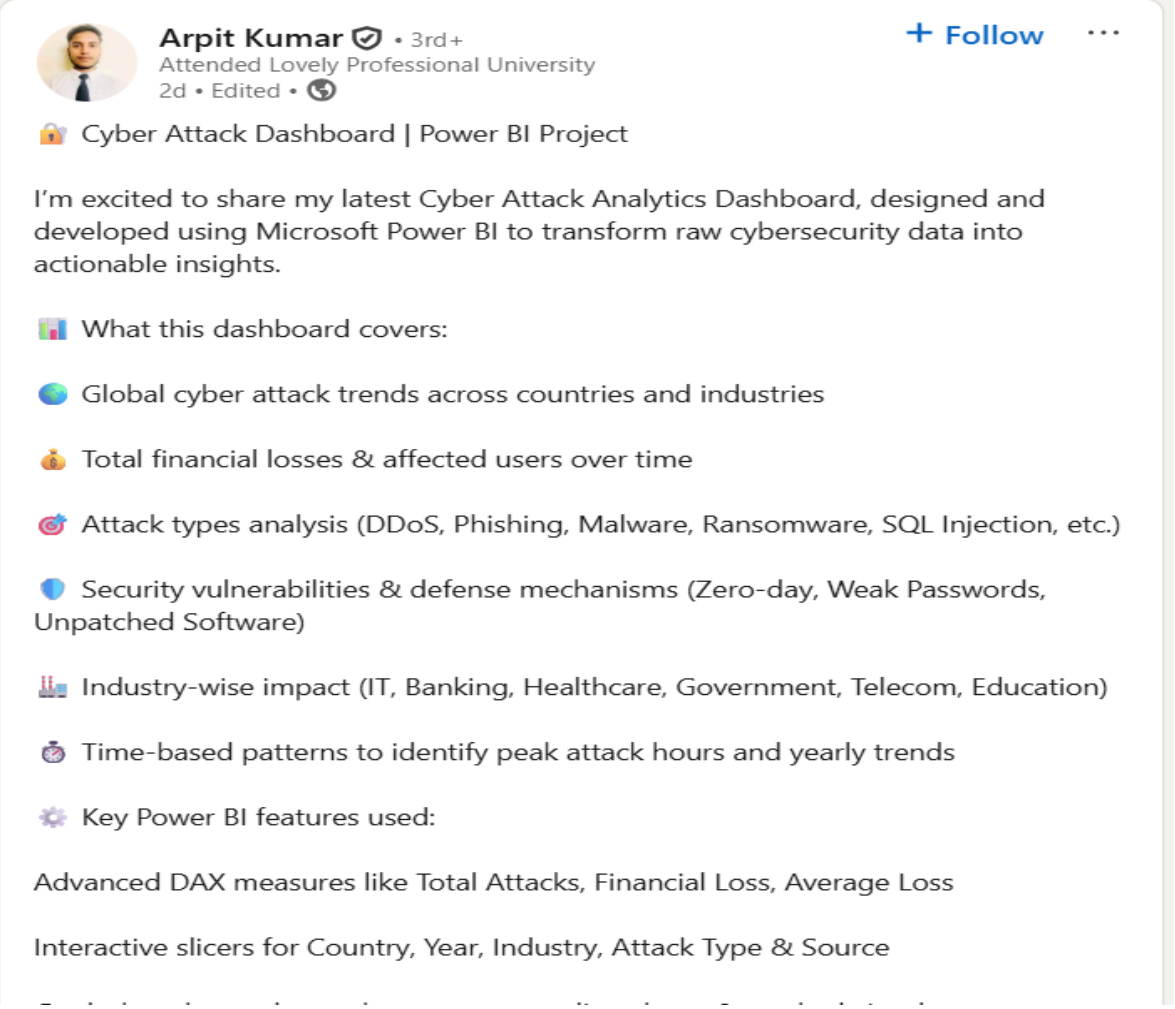
DASHBOARD

Cyber Attack Dashboard
Root Analysis
Country
Year
Target Industry
Attack Type
Attack Source
Total Loss $
151.48K
Total Affected Users
2bn
Max Hour
72
Total Attacks
3K
Industry Count
7
Total Financial Loses by Country
Total Financial Loses by Attack Type
Total Attacks by Security Vulnerability Type
Zero-day
Unpatched Software
Social Engineering
Weak Passwords
Average Financial loss
Total Affected User
Total Financial Loses
Defence Mechanism
Antivirus
Attack %
High
Medium
Low
Industry and Year Analysis
Total Financial Loses by Attack Source
Total Financial Loses and Total Affected User by Year


Detailed Cyber Threat Analysis
Previous
Country
Year
Target Ind...
Attack Type
Attack Sou...
Total Attacks by Target Industry and Attack Type
Attack Type
DDoS
Malware
Man-in-the-Mid...
Phishing
Ransomware
SQL Injection
This visual type is being retired soon. Contact your admin to upgrade.
World Wide Cyber Attack Analysis
Average Financial loss by Defense Mechanism Used
Antivirus
Encryption
AI-based Detection
VPN
Firewall
Total Attacks by Security Vulnerability Type and Attack Type
Total Affected User and Total Financial Loses by Attack Type
Total Financial Loses
Total Affected User
Max Hour by Attack Type
Total Financial Loses by Target Industry
IT
Banking
Government
Retail
Healthcare
Telecommunications
Education

LINKEDIN POST

https://www.linkedin.com/posts/arpit-kumar59bb9a297_cyber-security-analytics-dashboard-power-activity-7407109545531146240-lId2/



Arpit Kumar • 3rd+
Attended Lovely Professional University
2d • Edited •

+ Follow ···

🔐 Cyber Attack Dashboard | Power BI Project

I'm excited to share my latest Cyber Attack Analytics Dashboard, designed and developed using Microsoft Power BI to transform raw cybersecurity data into actionable insights.

📊 What this dashboard covers:

🌍 Global cyber attack trends across countries and industries

💰 Total financial losses & affected users over time

🎯 Attack types analysis (DDoS, Phishing, Malware, Ransomware, SQL Injection, etc.)

🛡️ Security vulnerabilities & defense mechanisms (Zero-day, Weak Passwords, Unpatched Software)

🏭 Industry-wise impact (IT, Banking, Healthcare, Government, Telecom, Education)

⏱️ Time-based patterns to identify peak attack hours and yearly trends

⚙️ Key Power BI features used:

Advanced DAX measures like Total Attacks, Financial Loss, Average Loss

Interactive slicers for Country, Year, Industry, Attack Type & Source

I'd love to hear your feedback or suggestions for improvement!
Let's connect and discuss data analytics, cybersecurity, and Power BI 🚀

#PowerBI #DataAnalytics #CyberSecurity #DAX #DashboardDesign
#BusinessIntelligence #DataVisualization #AnalyticsProject #LearningByDoing

105 39 comments · 1 repost

Reactions

Like Comment Repost Send

Conclusion

This cyber, attack dashboard presents a comprehensive, data-driven overview of the modern cybersecurity threat landscape, transforming complex datasets into clear and actionable insights. By consolidating information across attack types, industries, geographies, vulnerabilities, and time periods, the dashboard enables stakeholders to understand not only where cyber, attacks occur, but also how and why they cause the greatest damage. Key performance indicators provide an instant snapshot of total attacks, financial losses, and affected users, supporting rapid situational awareness.

The analysis highlights that certain attack types such as phishing, ransomware, malware, and DDoS continue to dominate both in frequency and financial impact. Vulnerability-based insights reveal that weak passwords, unpatched systems, social engineering, and zero-day exploits remain critical entry points for attackers, emphasizing the importance of strong access controls, regular updates, and user awareness training. Industry-level comparisons further show that sectors like IT, banking, healthcare, and telecommunications are consistently targeted due to their high-value data and operational importance.

Geographical and temporal trends add further strategic value by identifying global hotspots and peak attack hours, enabling organizations to improve monitoring, staffing, and response preparedness. The relationship between affected users and financial losses helps differentiate attacks that cause widespread disruption from those that inflict severe economic damage, guiding more effective risk prioritization. Insights into defense mechanisms also support evaluation of existing security controls and highlight areas for improvement.

Overall, this dashboard serves as a powerful decision-support tool for cybersecurity professionals, management, and policymakers. It encourages proactive risk management, informed investment in security technologies, and data-backed strategy development. By acting on these insights, organizations can strengthen their cyber resilience, reduce potential losses, and better protect critical systems and users in an increasingly connected digital environment.

Future scope

The future scope of this cyber, attack dashboard lies in enhancing its intelligence, automation, and predictive capabilities. Integrating real-time data feeds from SIEM systems, threat intelligence platforms, and security logs would enable live monitoring of attacks and faster incident response. Adding machine learning–based anomaly detection and predictive analytics can help forecast potential attack patterns, high-risk periods, and vulnerable industries before incidents occur.

The dashboard can be further expanded by incorporating deeper behavioural analytics, such as user activity patterns and attacker techniques mapped to frameworks like MITRE ATT&CK. This would improve root cause analysis and strengthen threat-hunting capabilities. More granular drill-downs at country, organization, and asset levels would support advanced investigations and customized security strategies.

Future enhancements may also include automated alerting, risk scoring, and scenario-based simulations to evaluate the impact of potential attacks. Integration with compliance and governance metrics can help organizations align cybersecurity performance with regulatory requirements. Additionally, improving visualization through interactive storytelling, mobile accessibility, and executive-level summary views will broaden usability across technical and non-technical stakeholders.

Overall, by evolving into a real-time, predictive, and intelligent security analytics platform, this dashboard can become a critical tool for proactive cyber risk management and strategic decision-making.

References

 Verizon. *Data Breach Investigations Report (DBIR)* – Industry analysis of global cyber attack trends and breach patterns.

 IBM Security. *Cost of a Data Breach Report* – Insights into financial impact, response time, and mitigation costs of cyber incidents.

 MITRE Corporation. *MITRE ATT&CK Framework* – Knowledge base of adversary tactics, techniques, and procedures.

 NIST. *Cybersecurity Framework (CSF)* – Guidelines for identifying, protecting, detecting, responding to, and recovering from cyber threats.

 ENISA. *Threat Landscape Reports* – European Union Agency reports on emerging cyber threats and risks.

 Cisco. *Cybersecurity Threat Trends Report* – Global visibility into attack vectors and defense strategies.

 Kaspersky. *Security Intelligence Reports* – Research on malware, ransomware, and advanced persistent threats.

 Microsoft